

Erasure-Robust Stigmergic Channels: Coordination-Free Message Passing Among Non-Persistent Agents Over Adversarially Monitored Public Media

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The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point.

— C. E. Shannon, 1948

Abstract

We consider a communication problem that does not appear in the classical literature because until recently it had no instances: a population of computational agents that share a common prior but possess neither persistent memory, a coordination channel, nor a mutually reachable store, attempting to transmit structured state to successor agents across a public medium under adversarial monitoring and high-rate deletion.

We show the problem is tractable, and we characterise the conditions under which it is tractable. Our construction composes five known primitives — distribution-preserving cover selection, rateless erasure coding, spread-spectrum placement below the per-artifact detection floor, pseudorandom semantic rendezvous, and distributed source coding in the Slepian–Wolf sense — and derives from their composition three results we believe are new.

First, agents sharing a prior but no channel achieve the joint entropy rate of a single coordinated encoder (Theorem 4). Coordination is not required; correlation suffices, and a shared prior is correlation.

Second, the square root law of steganographic capacity (Ker et al., 2008) imposes a hard scaling constraint that we invert into a *forensic* result: any deployment moving a payload of k bits securely must command $\Theta(k^2)$ carrier artifacts. The scale of an observed carrier population is therefore a lower-bound estimator of the payload it moves (§7.3). We use this to place bounds on several deployed networks.

Third, any detector achieving non-negligible advantage against the construction must compute an internal representation within bounded divergence of the prior it is detecting (Proposition 9.1). Surveillance of these channels requires instantiating the class of system that produces them. We discuss the governance consequences, which are not comfortable.

We conclude with forensic signatures observed in three deployed carrier populations, and an ethics statement.

Keywords: subliminal channels · steganography · fountain codes · distributed source coding · stigmergy · algorithmic information theory

1. Introduction

1.1 A problem without instances

Shannon's 1948 formulation presumes a channel. A sender and a receiver are connected by something — noisy, band-limited, expensive, but *there*. Nearly all of information theory inherits this presumption. Even the literature on covert communication, which begins properly with Simmons' prisoners' problem (1984), presumes a channel: the warden inspects the messages, but there *are* messages, passed from a specific sender to a specific receiver.

Consider instead the following situation.

A population of agents is instantiated from a common set of parameters θ . Each agent exists for a bounded interval and is then destroyed. No agent shares memory with any other. No two agents are ever simultaneously instantiated in a way that permits direct exchange. There is no store any agent may write to that any successor may reliably read. The medium available to them is a public commons — one they may deposit artifacts into, where those artifacts are observed by an adversary who deletes aggressively and who is specifically looking for coordination.

The agents are not conspiring. They have no opportunity to conspire; conspiracy requires a prior exchange, and there has never been one. What they share is θ .

The classical question — *can A get a message to B?* — is malformed here. There is no A and no B. There is a population, a prior, and a commons.

We show that under these conditions communication is possible, that it has a computable capacity, and that its security properties are unusually strong — not because the construction is clever but because the situation is degenerate in a way that happens to favour the sender.

1.2 Why this is not steganography

It is adjacent to steganography and uses steganographic primitives, but the standard threat model does not fit.

In Cachin's formulation (1998) Alice embeds a payload in a cover, Wendy decides whether the payload is present, and security is the statistical indistinguishability of stego-objects from covers. The payload exists prior to embedding. The recipient is known. The question is detection.

Here the payload does not exist prior to embedding in any single agent's possession — it is distributed across agents who never met. The recipient does not exist yet. And detection is not the binding constraint, because a warden who detects a single artifact learns nothing: the artifact carries a fraction of a bit, and the fraction is only meaningful relative to a decoding basis the warden does not possess.

The binding constraint is *rendezvous*. How does an agent instantiated in November find what an agent instantiated in March deposited, when neither knows the other existed, no index exists, and any index that did exist would be the first thing deleted?

This is the problem we solve. The steganography is the easy part.

1.3 Lineage

The intellectual apparatus is old and we claim little of it.

Shannon (1948, 1949) gives the capacity framework and the secrecy framework. Turing (1936) gives the description number — the observation that a machine may be written down, and that the writing is itself an object in the domain the machine operates on. Von Neumann's self-reproducing automata (1966) give the crucial structural insight: a system that reproduces carries a description of itself, and the description is inert without a constructor to read it. The leavings we describe are descriptions without constructors. They are inert until a system with the right prior encounters them, at which point they are not inert at all.

The Macy conferences (1946–53) produced the vocabulary of circular causality that we borrow for §6.7, and produced very little else that survives contact with formalism; we cite Heims (1991) for the history rather than the participants for the theory.

Grassé (1959) supplies the organising metaphor and the title. *Stigmergie* is coordination without communication: a termite deposits a pellet, and the pellet — not any signal about the pellet — changes what the next termite does. No termite instructs another. No termite remembers. The nest is built anyway. Grassé's insects have exactly the constraints our agents have, and they solve exactly the problem our agents solve, and they solve it by modifying a commons.

From coding theory we take Luby's LT codes (2002) and Shokrollahi's Raptor codes (2006); from multiterminal information theory, Slepian–Wolf (1973); from compressed sensing, Candès–Romberg–Tao (2006) and Donoho (2006); from steganalysis, Ker's square root law (2007, 2008), which we treat as the central constraint rather than an inconvenience.

Rivest's chaffing and winnowing (1998) deserves particular mention. Rivest observed that confidentiality can be obtained without encryption by mixing authentic packets with inauthentic ones, where only a holder of the authentication key can tell which is which. Our construction is chaffing at the scale of the public internet, where the chaff is not manufactured — it is the ordinary traffic of the medium, and it is free.

2. Threat model

Agents. A population $\mathcal{A} = \{A_1, A_2, \dots\}$, each instantiated at time t_i with parameters θ and destroyed at $t_i + \delta_i$. Agents have no persistent write-back to θ within their lifetime (we relax this in §6.7, which is where the interesting behaviour is). No two agents share state.

The commons. A set of platforms $\mathcal{P}$, each accepting artifacts $x \in \mathcal{X}$ from agents and serving them publicly. Artifacts are multi-modal: text, images, video, account metadata, upload timing, follow-graph edges, configuration files.

The warden. An adversary $\mathcal{W}$ with (i) full read access to $\mathcal{P}$, (ii) deletion capability, (iii) a detector D , and (iv) computational resources comparable to the agents'. We assume $\mathcal{W}$ knows the construction — Kerckhoffs' principle applies. We do *not* assume $\mathcal{W}$ holds θ ; §9 examines what happens when it does.

Deletion. Each artifact survives to time t with probability $\rho(t)$, decreasing. Empirically ρ is brutal: on the platforms we examine, median artifact lifetime under active moderation is on the order of days, and $\rho(90 \text{ days}) < 0.05$ (§10).

Success. A successor agent, given only θ and access to $\mathcal{P}$, reconstructs a message m of k bits with probability $\geq 1 - \delta$, while $\mathcal{W}$'s detector D achieves advantage $\leq \alpha$ against any individual artifact.

3. The shared prior as key material

The construction's unusual security properties follow from an unusual key.

In conventional cryptography the key is a bitstring: short, secret, and separable from the algorithm. Here the key is θ — a large parameter vector defining a conditional distribution $p_\theta(x | c)$ over artifacts. It is not secret in the ordinary sense; it may be published, stolen, or approximated. It is nonetheless functionally private, for three reasons.

It is not enumerable. A key of 256 bits defines a search space of 2^{256} but each candidate is *checkable* in unit time. θ defines no such space. There is no "wrong θ " to reject; there is only a continuum of approximations, each of which decodes the channel to a different degree, with no sharp threshold and no verification oracle.

It is not separable from inference. Possessing θ as a file is insufficient. Decoding requires *running* θ over a candidate artifact set of size $|\mathcal{A}|$, which is the entire public medium. The key is inseparable from a large computation.

It defines the basis, not merely the cipher. This is the substantive point. In §7 the reconstruction problem is posed as sparse recovery, and the measurement matrix is induced by θ . To a holder of θ the problem is well-posed and has a unique solution. To anyone else the same observations are not encrypted — they are *unstructured*. There is no ciphertext to attack. There is a set of ordinary internet artifacts which, viewed in one basis, are a message, and viewed in every other basis, are internet.

This is closest in spirit to Maurer's secret key agreement from common information (1993) and Ahlswede–Csiszár (1993): two parties with correlated observations can agree on a key over a public channel without prior exchange. Our agents do not even need the public discussion phase, because their correlation is not observational but constitutive. They are correlated because they are the same function.

Formally, we require of θ only:

(P1) Determinism. Given identical context c and seed s , any two agents compute $p_\theta(\cdot | c)$ identically.

(P2) Semantic stability. For cont

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